

Method / Paper	Venue / Year	What it does	Why it matters for RECA-FL
Multi-Task Learning Resource Allocation in Federated ISAC Networks	IEEE TWC 2024	Jointly optimizes sensing, communication, and computation resource allocation using multi-task learning / multi-objective optimization.	Very important baseline for ISAC-aware resource allocation without accommodation. (IEEE Xplore)
Integrated Sensing, Communication, and Computation for OTA-FL in 6G Wireless Networks	IEEE IoT-J 2024	Studies ISCC for over-the-air FL in 6G wireless networks.	Baseline for AirComp + ISAC + FL without world-model accommodation. (IEEE Xplore)
Over-the-Air FL Client Selection in Integrated Sensing, Computing and Communication	IEEE ICC Workshops 2024	Client selection for OTA-FL under integrated sensing/computing/communication.	Very close client-selection baseline. You should implement a simplified version if possible. (IEEE Xplore)
Communication-Efficient FL for Large-Scale Multiagent Systems in ISAC: Data Augmentation With RL	IEEE Systems Journal 2024	Uses reinforcement-learning-based data augmentation to handle non-IID in ISAC multiagent systems.	Important because your rare-regime/rare-class experiment overlaps with non-IID ISAC learning. (IEEE Xplore)
Integrated Sensing, Computation, and Communication for UAV-Assisted FEEL	IEEE TWC 2025	Joint UAV deployment and ISCC resource allocation for FEEL, with human motion recognition and probabilistic sensing.	Strong ISAC-FEEL baseline, especially if you include UAV/mobile-client setting. (IEEE Xplore)

Differentially Private Wireless FL With ISAC	IEEE TWC 2025	Jointly optimizes beamforming, device selection, resource allocation, and time scheduling under DP constraints.	Add either as a privacy-aware baseline or related work. If you do not include DP, cite it but do not force baseline comparison. (IEEE Xplore)
Integrated Sensing, Communication, and Computation for OTA Federated Edge Learning	IEEE TWC 2025/2026	Air-FEEL system with ISCC, sensing noise, AirComp distortion, energy/latency, and convergence analysis.	This is one of the closest baselines for your AirComp + sensing + learning setup. (IEEE Xplore)
Integrated Sensing, Computation, and Communication Enabled FEEL	IEEE TWC 2026	ISCC-enabled FEEL, resource-constrained wireless networks, ambient intelligence.	Very recent. Should be cited and ideally compared against if implementation is feasible. (IEEE Xplore)
Sensing-Native OTA-FL	arXiv 2026, IEEE-related authorship context	Reuses gradient signals and gradient statistics for zero-overhead sensing/localization in OTA-FL.	Not necessarily IEEE journal yet, but very recent and highly relevant. Add as related work and optional baseline. (arXiv)
Asaad-style OTA-FEEL with Integrated Sensing	2025 preprint	Joint scheduling and beamforming with CRB, MSE, and downlink rate constraints.	Extremely close to your sensing + AirComp selection story. Use as external baseline if codeable. (arXiv)

Method / Paper	Venue / Year	What it contributes	How to use it
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Agent Selection Framework for FL in Resource-Constrained Wireless Networks	IEEE TMLCN 2024	Defines an agent-selection metric using learning importance and wireless channel resource efficiency. It evaluates traffic-sign classification under latency/resource constraints.	Very good external baseline for client selection. Use it as “Learning + channel-aware selection.” (TU Delft Research Portal)
Toward Understanding FL over Unreliable Networks	IEEE TMLCN 2025	Studies convergence under unreliable wireless channels, scheduling policies, bandwidth, and momentum. Includes analytical convergence and empirical simulations.	Use as theory/style reference and maybe implement unreliable-channel scheduling baseline. (SUTD Repository)
Wireless Multimodal FL on Modal Heterogeneity	IEEE TMLCN 2025	Joint client scheduling and bandwidth allocation under latency, energy, and bandwidth constraints; validates on multimodal datasets.	Use as experimental-style reference: derive bound + optimize + validate on heterogeneous data. (SUTD Repository)
Personalized FL with Adaptive Transformer Pruning and Hypernetwork Personalization	IEEE TMLCN 2025/2026	Reports energy savings, training-time reduction, and accuracy under wireless-edge personalization.	Not central to ISAC, but useful if you include adapter/personalization overhead comparison. (Hamad Bin Khalifa University)
Decentralized Aggregation / FedMoD-style D2D FL	IEEE TMLCN 2024	Decentralized/energy-efficient FL over aerial-terrestrial or D2D-style wireless settings.	Optional. Use if you discuss decentralized aggregation or energy efficiency. (IEEE Xplore)